CASE STUDY 42

STUDENT MARKS MATRIX ANALYSIS

Python Program:

```
print("===== STUDENT MARKS MATRIX ANALYSIS =====")


students = int(input("Enter number of students: "))
subjects = int(input("Enter number of subjects: "))


if students <= 0 or subjects <= 0:
    print("Error: Number of students and subjects must be positive.")


else:
    marks = []


    for i in range(students):
        student_marks = []
        print("\nEnter marks for Student", i + 1)


        for j in range(subjects):
            mark = float(
                input("Enter marks for Subject {}: ".format(j + 1))
            )


            if mark < 0 or mark > 100:
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            print("Invalid marks! Enter marks between 0 and 100.")
            break

        student_marks.append(mark)

    if len(student_marks) != subjects:
        break

    marks.append(student_marks)

if len(marks) == students:
    print("\n===== MARKS MATRIX =====")

    for row in marks:
        print(row)

    print("\n===== STUDENT-WISE ANALYSIS =====")

    for i in range(students):
        total = sum(marks[i])
        average = total / subjects

        print(
            "Student {}: Total = {:.2f}, Average = {:.2f}".format(
                i + 1, total, average
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        )
    )

print("\n===== SUBJECT-WISE ANALYSIS =====")

for j in range(subjects):
    subject_marks = []

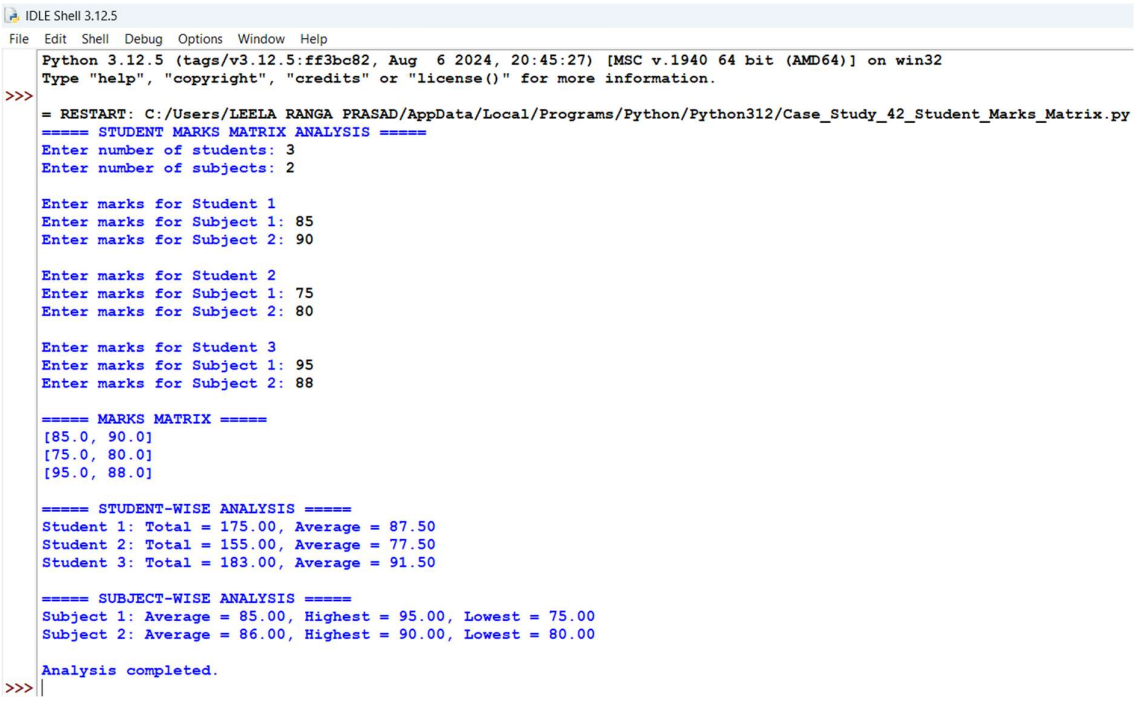
    for i in range(students):
        subject_marks.append(marks[i][j])

    average = sum(subject_marks) / students

    print(
        "Subject {}: Average = {:.2f}, Highest = {:.2f}, "
        "Lowest = {:.2f}".format(
            j + 1,
            average,
            max(subject_marks),
            min(subject_marks)
        )
    )

print("\nAnalysis completed.")
```

Output:


IDLE Shell 3.12.5
File Edit Shell Debug Options Window Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/LEELA RANGA PRASAD/AppData/Local/Programs/Python/Python312/Case_Study_42_Student_Marks_Matrix.py
===== STUDENT MARKS MATRIX ANALYSIS =====
Enter number of students: 3
Enter number of subjects: 2
Enter marks for Student 1
Enter marks for Subject 1: 85
Enter marks for Subject 2: 90
Enter marks for Student 2
Enter marks for Subject 1: 75
Enter marks for Subject 2: 80
Enter marks for Student 3
Enter marks for Subject 1: 95
Enter marks for Subject 2: 88
===== MARKS MATRIX =====
[85.0, 90.0]
[75.0, 80.0]
[95.0, 88.0]
===== STUDENT-WISE ANALYSIS =====
Student 1: Total = 175.00, Average = 87.50
Student 2: Total = 155.00, Average = 77.50
Student 3: Total = 183.00, Average = 91.50
===== SUBJECT-WISE ANALYSIS =====
Subject 1: Average = 85.00, Highest = 95.00, Lowest = 75.00
Subject 2: Average = 86.00, Highest = 90.00, Lowest = 80.00
Analysis completed.
>>>